Agentic AI for Oral Cancer Screening

A Hierarchical Multi-Agent System Integrated with a Smart Toothbrush IoT Device

Project	**System Type**	**Scope**
Integrated Design Project — Part 2	Hierarchical Agentic AI + IoT	Concept architecture + interaction design

Executive Summary

Oral cancer is among the most survivable cancers when detected early, yet the majority of cases are diagnosed at advanced stages because routine screening depends on clinic visits most people never make. This project proposes an agentic AI system embedded into a daily routine — brushing teeth — to move screening from episodic to continuous, and from reactive to proactive.

The system is built as six cooperating AI agents arranged in a hierarchy beneath a central Orchestrator. A smart toothbrush with an embedded micro-camera and motion sensors feeds raw data to perception agents, which hand off to reasoning agents that produce two distinct outputs: a plain-language report for the patient, and a clinical referral packet for a dentist or oncologist. Crucially, the AI does not diagnose — it flags, explains, and routes.

This document describes the system from three perspectives: the patient using the device (user view), the clinician receiving referrals (clinician view), and the internal agent architecture (logic view). It also defines agent responsibilities, data flow, a technology stack suitable for a prototype, and known limitations.

1. Problem Framing

1.1 Clinical Gap

Oral cavity cancers (lip, tongue, floor-of-mouth, buccal mucosa) are strongly linked to tobacco, alcohol, and betel-quid use — risk factors prevalent across South and Southeast Asia. Five-year survival exceeds 80% when lesions are caught at stage I but drops below 40% at stage IV. The gap between these two numbers is almost entirely a screening problem: visible precancerous lesions (leukoplakia, erythroplakia) can be detected months or years before malignancy, but only if someone looks.

1.2 Why Current Screening Fails

Dental visits are infrequent, opportunistic, and skewed toward populations already engaged with healthcare. Self-examination guides exist but are rarely followed. Telemedicine approaches require the patient to take the initiative — a high bar for an asymptomatic condition. The screening behaviour needs to be invisible, automatic, and embedded in something people already do.

1.3 Design Principle

The toothbrush is the ideal substrate: used twice a day by most adults, pointed directly at the screening target, and already a consumer product people purchase without a medical prompt. Adding a camera and pairing it with an agentic AI backend turns every brushing session into a passive screening event. The user does not need to remember to screen — they just need to keep brushing.

2. System Overview — Three Views

This system is described from three perspectives because its success depends on each being coherent: the patient must find it effortless, the clinician must find it trustworthy, and the underlying logic must be defensible. The three diagrams below are not redundant views of the same thing — they are three different contracts the system must fulfil simultaneously.

2.1 User View — The Patient Journey

From the patient's side, the system is almost invisible. They brush their teeth. The brush captures imagery of the oral cavity during the two-minute session. By the time they have rinsed and put the brush down, the app has a result waiting. Most days the result is a green badge and a short tip on brushing technique. On the rare day the system flags something, the message is specific: a region, a reason, and a next step that has already been arranged.

Figure 1 — Patient Journey (User View)

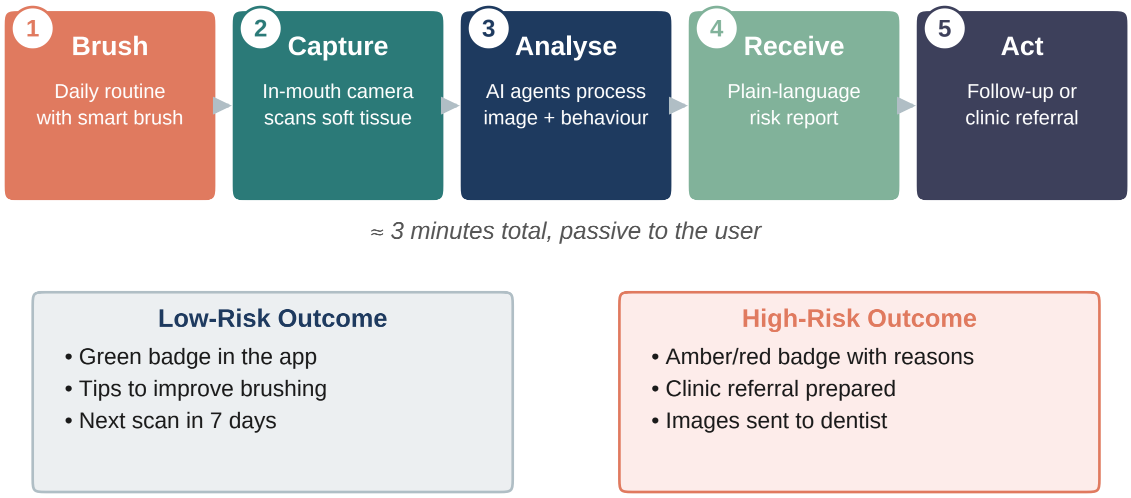

Figure 1 · The patient-facing flow. The user does not interact with the AI directly — they interact with a result, and only when the result is actionable.

2.2 Clinician View — The Handoff

From the clinician's side, the system is a triage pipeline, not a diagnostic oracle. When a case is referred, the dentist or oncologist receives a structured packet: the suspect images with a heat-map overlay, the numerical risk score with its top contributing factors, thirty days of brushing history, and the patient's self-reported risk questionnaire. The clinician makes the diagnostic call; the system makes the case worth looking at.

Every clinician decision — confirm, reject, reclassify — is captured as a labelled data point and fed back into the evaluation set for the Screening Agent's vision model. This is a review loop, not an auto-retraining loop: retraining is always manual, versioned, and signed off before deployment.

Figure 2 — Clinician Handoff (Dentist / Oncologist View)

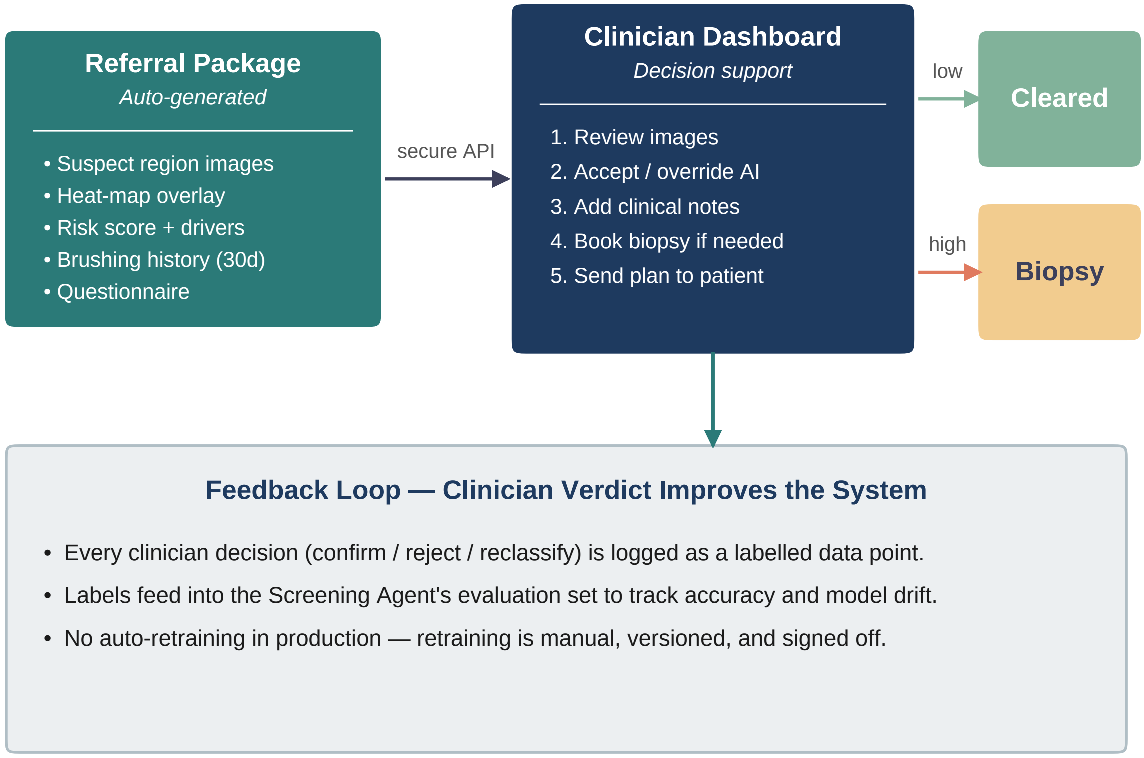

Figure 2 · The clinician-facing handoff. Note that the AI's output is evidence, not a verdict — final clinical judgement stays with the human.

2.3 Logic View — The Agent Architecture

Beneath the two user-facing views is the actual machinery: six AI agents arranged in four layers. The architecture is hierarchical — one Orchestrator Agent routes work and manages state, while the other five agents are specialists invoked on demand. This structure is deliberate. A flat peer-to-peer design would be more symmetric but would scatter decision authority, making it harder to audit why a particular referral was generated. A strict pipeline would be auditable but rigid. Hierarchy preserves both clarity and flexibility.

Figure 3 — Internal Logic (Hierarchical Agent Architecture)

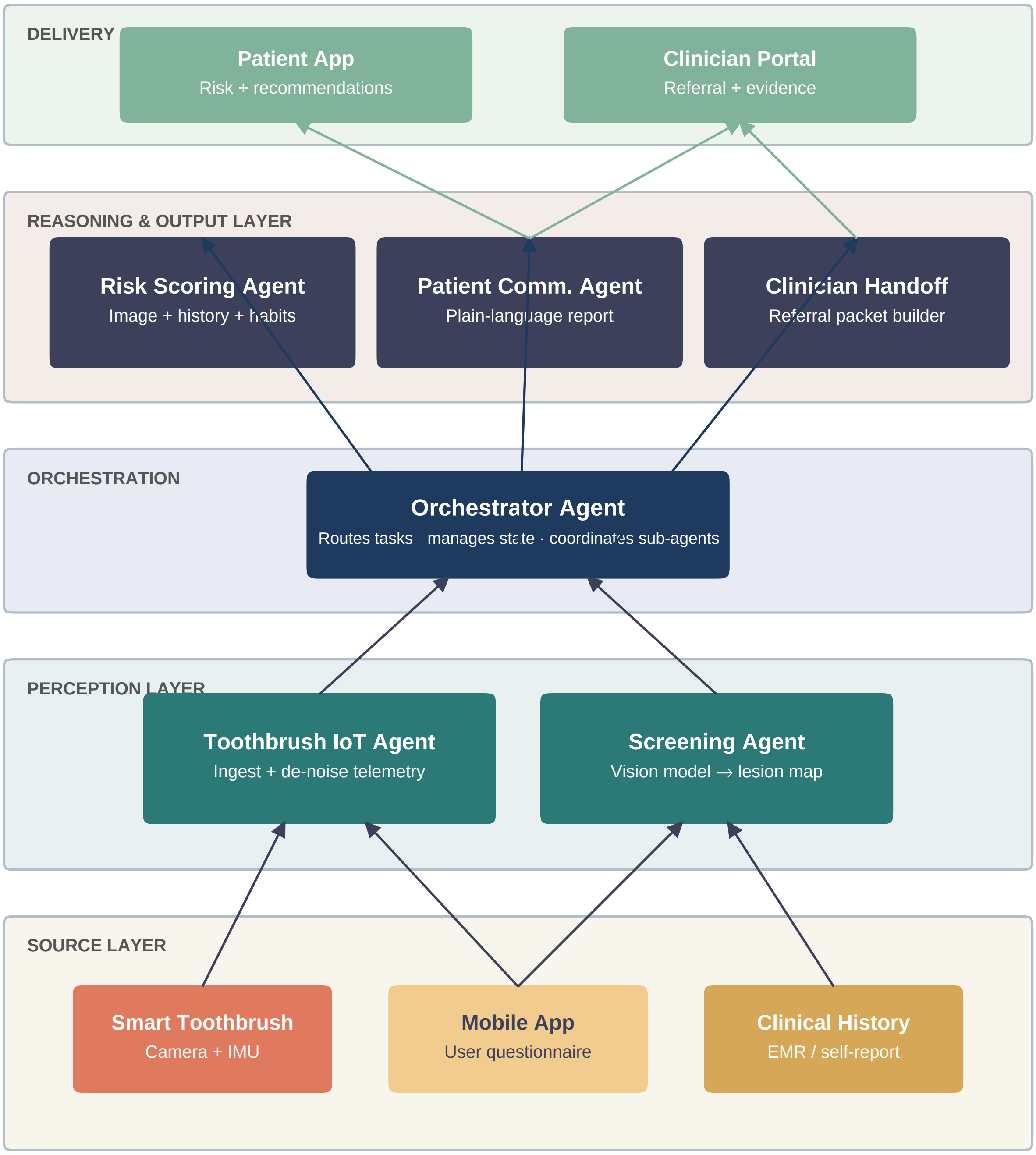

Figure 3 · The internal multi-agent architecture. Data flows upward from the Source Layer through Perception into the Orchestrator, which dispatches work to the Reasoning agents and routes their outputs to the appropriate Delivery channel.

3. Agent Responsibilities

Each of the six agents has a single, well-defined responsibility. The Orchestrator Agent owns workflow but has no domain reasoning of its own; the specialist agents own domain reasoning but have no knowledge of each other. This separation is what allows an agent to be swapped, upgraded, or audited in isolation.

Agent	Layer	Input	Output	Owns
Orchestrator	Coord.	Events from any source	Task dispatches, state updates	Workflow, routing, state
Toothbrush IoT	Perception	Raw telemetry (camera frames, pressure, IMU)	Cleaned frames + brushing metrics	Sensor data ingestion
Screening	Perception	Cleaned oral-cavity frames	Lesion map, suspect regions, per-region confidence	Vision model inference
Risk Scoring	Reasoning	Lesion output + questionnaire + history	Composite risk score + top drivers	Risk synthesis logic
Patient Comm.	Reasoning	Risk score + context	Plain-language message + next-step card	Tone, clarity, safety
Clinician Handoff	Reasoning	Risk score (when high) + evidence	Structured referral packet + API call	Referral formatting, integrations

Table 1 · Agent responsibilities. Each row defines a contract: given this input, the agent guarantees that output. The Orchestrator is the only agent aware of the others.

3.1 The Orchestrator's Role

The Orchestrator Agent is not a decision-maker. It is a dispatcher. When a brushing session completes, the Orchestrator receives an event, invokes the Toothbrush IoT Agent to clean the telemetry, passes the output to the Screening Agent, and — once the lesion map is back — calls the Risk Scoring Agent with the combined context. Based on the resulting score, the Orchestrator chooses which downstream reasoning agents to invoke: always the Patient Communication Agent, and only above a risk threshold, the Clinician Handoff Agent. This single point of routing is also the single point of audit: every case has one trace.

3.2 Why Not a Single Large Model?

A single large model could, in principle, take raw frames and emit a referral packet. The reason this system does not is accountability. A monolithic model is a black box whose failure modes are entangled. A six-agent system with explicit interfaces lets us isolate the question "did this referral get generated because the vision model was wrong, or because the risk scoring was wrong?" — a question clinicians and regulators will ask, and that we need to be able to answer.

4. Data Flow — A Worked Example

To make the architecture concrete, this section walks through a single brushing session end-to-end, showing which agent holds what data at each step.

4.1 Session Start

The patient picks up the brush. An accelerometer-triggered wake event is sent over BLE to the mobile app, which opens a session and notifies the Orchestrator Agent (over MQTT). The Orchestrator creates a session ID and a state record in the backend database.

4.2 Capture and Ingestion

During the two-minute session, the brush streams camera frames (throttled to 2 fps for bandwidth) and IMU readings to the app. The app forwards them to the Toothbrush IoT Agent, which filters blurred frames, aligns timestamps, and summarises the brushing behaviour (coverage by quadrant, pressure distribution, total time).

4.3 Screening Inference

The cleaned frames are passed to the Screening Agent, which runs a vision model fine-tuned on oral-lesion imagery. The output is a per-frame lesion map: bounding regions with classification probabilities (healthy, benign lesion, suspect lesion) and a confidence score. Low-confidence frames are discarded; the highest-confidence suspect regions are promoted for downstream reasoning.

4.4 Risk Synthesis

The Orchestrator now has two streams of evidence (image-based and behavioural) and invokes the Risk Scoring Agent, also providing the patient's questionnaire responses (tobacco, alcohol, betel-quid use, age, family history). The agent produces a composite risk score on a fixed scale along with the top contributing factors — critical for explainability.

4.5 Output Routing

The Orchestrator always invokes the Patient Communication Agent, which translates the score and drivers into a short, plain-language message. If the score exceeds a referral threshold, the Orchestrator additionally invokes the Clinician Handoff Agent, which packages the evidence and sends it (via an authenticated API) to the patient's registered clinic. The referral status is written back to the patient app so the user sees what was sent and to whom.

4.6 Session Close

The Orchestrator closes the session record, retaining the image references, risk score, and routing decisions for audit. Raw frames themselves are retained only with explicit consent, in encrypted storage, with a fixed retention window.

5. Technology Stack

The prototype stack prioritises interfaces over components — every piece can be swapped as long as the agent contracts in Table 1 are preserved.

Layer	Component	Choice (prototype)
Hardware	Toothbrush device	Embedded micro-camera + 6-axis IMU + pressure sensor

Transport	Device → cloud	BLE device → phone, MQTT phone → cloud
Agents	Reasoning (Orchestrator, Risk, Comm., Handoff)	Claude API (reasoning + tool use)
Agents	Perception (Screening)	Fine-tuned CNN / vision transformer on oral lesion data
Agents	Perception (IoT)	Python service, numpy / opencv for frame cleaning
Backend	API + orchestration	FastAPI (Python), async task queue
Data	Session + agent state	Supabase (Postgres + auth + storage)
Delivery	Patient app	React Native, consumes FastAPI
Delivery	Clinician portal	Next.js web app, role-gated access

Table 2 · Prototype technology stack.

6. Evaluation Plan

The system has two kinds of correctness: **technical** (does each agent do its job accurately?) and **clinical** (does the system as a whole help patients reach care earlier?). Both need separate evaluation.

6.1 Agent-Level Metrics

The Screening Agent is evaluated against a held-out labelled set of oral-lesion images using standard metrics (sensitivity, specificity, AUROC). The Risk Scoring Agent is evaluated against retrospective case data — would it have flagged the cases that were eventually diagnosed? The Patient Communication Agent is evaluated via readability scores and user-comprehension tests, not accuracy alone. The Clinician Handoff Agent is evaluated on referral completeness: does the packet contain everything the clinician asked for?

6.2 System-Level Metrics

At the system level, the primary metric is **time-to-clinical-contact** for flagged cases — the hypothesis is that embedded screening shortens the interval between a detectable lesion appearing and a clinician examining it. Secondary metrics include user retention (do people keep using the brush?), referral acceptance rate (do clinicians find the referrals useful?), and false-positive fatigue (do users start ignoring alerts?).

7. Limitations and Risks

7.1 False Negatives Are Worse Than False Positives

A missed lesion is a direct harm; a false alarm is an inconvenience and a clinic visit. The Risk Scoring threshold is therefore deliberately tuned toward over-referral rather than under-referral. This is an explicit design choice, not a bug, and clinicians should be briefed on it so they understand why they see low-yield referrals.

7.2 Image Quality at the Hardware Edge

A toothbrush camera is not a clinical camera. Lighting, angle, saliva, and motion blur will degrade input quality. The Toothbrush IoT Agent filters aggressively, but sessions with insufficient usable frames must be handled gracefully (no result rather than a wrong result).

7.3 Population Bias in Training Data

Publicly available oral lesion datasets skew toward certain demographics and clinical settings. A model trained on them may underperform on under-represented groups, including the betel-quid-using populations this system is most aimed at. Validation on representative cohorts is a prerequisite to deployment, not an afterthought.

7.4 Privacy and Consent

Capturing imagery inside the human body is intimate data. The system must treat image retention, sharing, and deletion as first-class design concerns — not as compliance boxes. Default behaviour should favour minimal retention and explicit consent for every data-sharing event.

7.5 Scope Discipline

The system is a screening aid. It does not diagnose, does not prescribe, and does not replace a dentist. Every user-facing message and every clinician-facing packet should reinforce this, not because of liability theatre, but because stepping outside this scope is where medical AI systems cause real harm.

8. Future Work

Several extensions are deferred from the current scope but worth naming:

- **Insurance Partner Agent.** A seventh agent that, with patient consent, communicates screening adherence to preventive-care insurance products, potentially reducing premiums for engaged users. Intentionally out of scope for the MVP to keep the system focused on clinical correctness before commercial pathways.
- **Multi-modal expansion.** Current perception uses image and motion telemetry. Breath-based volatile organic compound sensing and saliva biomarker assays are emerging adjuncts that could be added as new Perception-layer agents without changing the rest of the architecture.
- **Longitudinal tracking.** The most powerful signal in screening is often not a single image but the change across images over time. A dedicated temporal-analysis agent could track individual regions across months.
- **Offline-first clinician portal.** For rural deployments, a version of the clinician portal that works on intermittent connectivity and syncs when available.

9. Conclusion

Oral cancer is a screening problem more than a treatment problem, and screening is a behavioural problem more than a technical one. Embedding an agentic AI system into the one oral-health behaviour people already perform daily collapses the friction that keeps most precancerous lesions from ever being seen by a clinician.

The six-agent hierarchical architecture presented here is designed around three properties: it is invisible to the patient, trustworthy to the clinician, and auditable to the regulator. Each agent has a single responsibility, a single contract, and a single path of accountability through the Orchestrator. What the system trades for this clarity is monolithic intelligence — but the ability to point at the exact agent responsible for any given decision is, for a medical system, more valuable than raw capability.

The next milestone is a working prototype of the perception layer — Toothbrush IoT and Screening agents — connected to a minimal Orchestrator and a mock clinician portal. Once that loop closes end-to-end on a single session, the reasoning agents can be layered on with confidence that the foundation holds.